Christo Aluckal

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About Me

PhD candidate in Computer and Information Science developing autonomy and perception systems for real-world robots. My research focuses on robot navigation, simulation environments, and scalable 3D perception pipelines, combining real-robot deployment with simulation-driven experimentation.
Research Interests: Embodied AI, robot learning, simulation-to-real transfer, robot navigation and perception, 3D scene reconstruction
Core Technologies: ROS2 • Nav2 • LiDAR Perception • Robot Simulation • Unity • Gazebo

Experience

- **DRONESLab** — *Oct 2022- Present*
  Graduate Research Assistant / PhD Candidate — Buffalo, United States
  - Implemented a **ROS2 Nav2** navigation stack on the **Unitree GO1** quadruped, designing **custom layered costmaps** that fuse 3D LiDAR perception with proprioceptive state information for robust real-time obstacle avoidance during autonomous navigation.
  - **Developed TERA**, a **ROS2-integrated Unity3D** simulation framework for autonomous excavators supporting configurable robot hardware and sensor suites (**LiDAR, cameras, IMU**) for autonomy algorithm testing and evaluation.
  - Developed a **URDF robot model and ROS2 control interface** for an excavator platform in **Gazebo** Classic and Ignition, integrating high-level autonomy algorithms with low-level joint control for autonomous excavation experiments.
  - **Designed an incremental training strategy for 3D Gaussian Splatting** enabling memory-efficient real-time scene reconstruction on edge and low-power devices, **reducing training time by 33%**.
  - Contributed to perception software for **autonomous inspection in low-light environments** (e.g., culverts), supporting robotic infrastructure inspection workflows.
- **InfiCorridor Solutions Pvt. Ltd.** — *May 2020 — Jul 2022*
  Intern / Part-Time — Mumbai, India
  - Developed **PixTrigger**, an open-source Python/C++ framework for synchronized image capture and geotagging supporting **100+ camera systems** for UAV inspection and mapping workflows.
  - Implemented the **NPNT (No Permission No Take-off) system** in C/C++, verifying **digital signatures** to enforce aviation authority regulations and ensure only authorized UAV missions execute.
  - Planned and executed **autonomous UAV missions** for terrain mapping and inspection, generating geospatial datasets used to produce **Digital Elevation and Terrain Models**.
  - Performed data auditing, validation, and verification pipelines for **geospatial datasets** used in GIS applications.

Projects

- **Master's Thesis: TERA – Robotics Simulation Framework for Excavation Autonomy**
  Tools: [Unity3D, C#, ROS2]
  - Designed a **robotics simulation environment** supporting real-time physics and high-fidelity modeling of excavator-terrain interaction for evaluating **excavation autonomy** algorithms.
  - Incorporated time-varying dynamics to study the challenges of full excavation autonomy.
  - Implemented configurable sensor spawning (**cameras, IMUs, LiDAR**) with customizable parameters to support perception and autonomy research.
  - Validated simulation fidelity through experiments including track deformation, velocity control profiles, and **sim-to-real comparison studies**.
- **OpenVINS Gaussian Splatting**
  Tools: [Python, PyTorch, CUDA, ROS2, OpenVINS, 3D Gaussian Splatting]
  - Built a replay-first visual-inertial mapping pipeline connecting ROS2/OpenVINS packet exports to 3D Gaussian Splatting, training directly from packets without requiring COLMAP-formatted scenes at runtime.
  - Implemented packet-backed fixed-pose scene loading with calibrated intrinsics, covariance, sparse feature tracks, and per-frame image metadata for 640x640 **TartanAir/OpenVINS** reconstruction sequences.
  - Integrated EDGS/RoMa correspondence initialization and LoD training modes, supporting 4 controlled experiment variants: vanilla 3DGS, EDGS, LoD, and EDGS+LoD.

- Reduced final Gaussian count by 50.1% and end-to-end training time by 21.6% on 50,000-iteration reconstruction runs while maintaining image quality at 26.17 dB PSNR.

- **PIXER: Enhancing Visual Odometry with Reliable Pixel Masking**
  Tools: [Python, BNNs, PySLAM, Optuna] [🌐]
  - Evaluated PIXER, a probabilistic feature-selection method for **visual odometry**, improving keypoint efficiency while maintaining localization accuracy.
  - Collected a custom dataset using **ZED 2i camera + Mosaic X5 GNSS** mounted on a **Boston Dynamics Spot** robot.
  - Integrated PIXER outputs into **PySLAM** and implemented **Optuna-driven hyperparameter optimization** with parallel studies to benchmark accuracy, runtime, and feature efficiency.
  - Achieved **34% RMSE reduction** and **41–49% keypoint reduction** in visual odometry experiments, enabling faster VO using lightweight feature detectors.

Education

- **University at Buffalo** — *Jan 2025 - Present*
  Doctor of Philosophy, Computer and Information Science — Buffalo, United States
- **University at Buffalo** — *Aug 2022 - Jan 2025*
  Master of Science, Computer and Information Science — Buffalo, United States
  - GPA: 3.53/4.00
  - Relevant Courses: Machine Learning, Reinforcement Learning, Robotics Algorithms, Computer Vision & Image Processing, Learning for Autonomous Systems
- **Mumbai University** — *Aug 2016 - Oct 2021*
  Bachelor of Engineering, Computer Engineering — Mumbai, India
  - Grade: 8.44/10.00
  - Relevant Courses: Machine Learning, Digital Signal & Image Processing

Publications

C=Conference, S=In Submission, W=Workshop, Po=Poster

[C.1] C. Aluckal et al (2025). **"TERA: A Simulation Environment for Terrain Excavation Robot Autonomy,"**. In *2025 IEEE International Conference on Simulation, Modeling, and Programming for Autonomous Robots (SIMPAR), Palermo, Italy, 2025, pp. 1-6, doi: 10.1109/SIMPAR62925.2025.10979147*

[C.2] Y. Turkar, T. Chase Jr, **C. Aluckal**, and K. Dantu, **PIXER: Enhancing Visual Odometry with Reliable Pixel Masking**. In in 23rd Conference on Robots and Vision, 2026

[W.1] Y. Turkar, **C. Aluckal**, S. Sreedharan, Y. Dighe, Y. Kim, J. Gemerek, and K. Dantu (2025) **Excavation autonomy with resilient traversability and handling**. In *Workshop on Field Robotics (WFR), International Conference on Robotics and Automation (ICRA) 2025*

[C.3] Yash Turkar; **Christo Aluckal**; Shaunak De; Varsha Turkar; Yogesh Agarwadkar (2022). **Generative-Network Based Multimedia Super-Resolution for Uav Remote Sensing**. In *IGARSS 2022 - 2022 IEEE International Geoscience and Remote Sensing Symposium, Kuala Lumpur, Malaysia, 2022, pp. 527-530, doi: 10.1109/IGARSS46834.2022.9884486.*

[C.4] Y. Turkar, **C. Aluckal**, C. Adhivarahan, A. Sebastiani, and K. Dantu, **A view-planning approach to 3d reconstruction**. In *International Conference on Extended Reality, Springer, 2024, pp. 340–350*

[Po.1] C. Aluckal, C. Adhivarahan, K. Dantu. **Splat-by-Splat: Progressive Gaussian Splatting for Efficient Scene Reconstruction**. In *NERC 2025, Cornell, Ithaca*

Skills

- **Programming Languages:** Python, C, C++, C#, OCaml
- **Robotics & Autonomy:** ROS / ROS2, Nav2, URDF, ros2_control, motion planning, robot navigation, LiDAR perception, robot perception pipelines, sim-to-real experimentation
- **Machine Learning & 3D Perception:** PyTorch, TensorFlow, 3D Gaussian Splatting, OpenCV, NumPy, Pandas, Optuna, WandB
- **Simulation & Robotics Platforms:** Gazebo (Classic & Ignition), Unity3D, NVIDIA Isaac Sim, ROS2 simulation ecosystems
- **Systems & Development Tools:** Linux, Git, LaTeX

Awards

- **Just Joe Sportsmanship Award** — *June 2019*
  Student Unmanned Aerial Systems Competition — [Page 20 🌐]
- **CSE Faculty Choice Award** — *May 2025*
  University at Buffalo